

Algebra: Inverse of a Function

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DIRECTIONS

Find the inverse function $f^{-1}(x)$ for each. Replace $f(x)$ with y , swap x and y , then solve for y .

1. Find $f^{-1}(x)$:

$$f(x) = 3x + 6$$

Answer: _____

2. Find $f^{-1}(x)$:

$$f(x) = 2x - 7$$

Answer: _____

3. Find $f^{-1}(x)$:

$$f(x) = -2x + 5$$

Answer: _____

4. Find $f^{-1}(x)$:

$$f(x) = 4x - 3$$

Answer: _____

5. Find $f^{-1}(x)$:

$$f(x) = x + 8$$

Answer: _____

6. Find $f^{-1}(x)$:

$$f(x) = -x + 4$$

Answer: _____

7. Find $f^{-1}(x)$:

$$f(x) = \frac{x - 2}{5}$$

Answer: _____

8. Find $f^{-1}(x)$:

$$f(x) = \frac{x + 1}{3}$$

Answer: _____

9. Find $f^{-1}(x)$:

$$f(x) = x^3 + 1$$

Answer: _____

10. If $f(3) = 7$, find:

$$f^{-1}(7) = ?$$

Answer: _____

Based on the Numberbender lesson "How to Find the Inverse of a Function, Algebraically and Graphically" •

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Answer Key & Solutions

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TEACHER NOTES

Items 1-6: linear functions — swap x and y then isolate. Items 7-8: fractional linear — multiply to clear denominator. Item 9: cube root reverses a cubic. Item 10: conceptual — if $f(a) = b$ then $f^{-1}(b) = a$.

1. Find $f^{-1}(x)$: $f(x) = 3x + 6$

Answer: $f^{-1}(x) = (x - 6) / 3$

$$y = 3x+6 \rightarrow x = 3y+6 \rightarrow y = (x-6)/3.$$

2. Find $f^{-1}(x)$: $f(x) = 2x - 7$

Answer: $f^{-1}(x) = (x + 7) / 2$

$$x = 2y-7 \rightarrow 2y = x+7 \rightarrow y = (x+7)/2.$$

3. Find $f^{-1}(x)$: $f(x) = -2x + 5$

Answer: $f^{-1}(x) = (5 - x) / 2$

$$x = -2y+5 \rightarrow 2y = 5-x \rightarrow y = (5-x)/2.$$

4. Find $f^{-1}(x)$: $f(x) = 4x - 3$

Answer: $f^{-1}(x) = (x + 3) / 4$

$$x = 4y-3 \rightarrow 4y = x+3 \rightarrow y = (x+3)/4.$$

5. Find $f^{-1}(x)$: $f(x) = x + 8$

Answer: $f^{-1}(x) = x - 8$

$$x = y+8 \rightarrow y = x-8.$$

6. Find $f^{-1}(x)$: $f(x) = -x + 4$

Answer: $f^{-1}(x) = 4 - x$

$$x = -y+4 \rightarrow y = 4-x. \text{ This function is its own inverse!}$$

7. Find $f^{-1}(x)$: $f(x) = (x - 2) / 5$

Answer: $f^{-1}(x) = 5x + 2$

$$x = (y-2)/5 \rightarrow 5x = y-2 \rightarrow y = 5x+2.$$

8. Find $f^{-1}(x)$: $f(x) = (x + 1) / 3$

Answer: $f^{-1}(x) = 3x - 1$

$$x = (y+1)/3 \rightarrow 3x = y+1 \rightarrow y = 3x-1.$$

9. Find $f^{-1}(x)$: $f(x) = x^3 + 1$

Answer: $f^{-1}(x) = \sqrt[3]{x - 1}$

$$x = y^3+1 \rightarrow y^3 = x-1 \rightarrow y = \sqrt[3]{x-1}.$$

10. If $f(3) = 7$, find: $f^{-1}(7) = ?$

Answer: $f^{-1}(7) = 3$

$$\text{If } f(a) = b, \text{ then } f^{-1}(b) = a. \text{ Since } f(3) = 7, f^{-1}(7) = 3.$$

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